

Chain Rule

Q. 1 Let $z = \ln(x^2 + 1)$ and $x = r \cos \theta$ Then find $\frac{\delta z}{\delta r}$ and $\frac{\delta z}{\delta \theta}$

Solution:

$$\begin{aligned}\frac{\delta z}{\delta r} &= \frac{\delta z}{\delta x} \cdot \frac{\delta x}{\delta r} \\&= \frac{\delta}{\delta x}(\ln(x^2 + 1)) \cdot \frac{\delta}{\delta r}(r \cos \theta) \\&= \frac{1}{x^2 + 1} \frac{d}{dx}(x^2 + 1) \cdot \cos \theta \\&= \frac{2x}{x^2 + 1} \cdot \cos \theta \\&= \frac{2r \cos \theta \cdot \cos \theta}{r^2 \cos^2 + 1}\end{aligned}$$

Now,

$$\begin{aligned}\frac{\delta z}{\delta \theta} &= \frac{\delta z}{\delta x} \cdot \frac{\delta x}{\delta \theta} \\&= \frac{\delta}{\delta x}(\ln(x^2 + 1)) \cdot \frac{\delta}{\delta \theta}(r \cos \theta) \\&= \frac{2x}{x^2 + 1} (-r \sin \theta) \\&= -\frac{2r \cos \theta}{(r \cos \theta)^2 + 1} r \sin \theta \\&= \frac{2r^2 \cos \theta \sin \theta}{r^2 \cos^2 \theta + 1} \\&= \frac{2r^2 \sin \theta \cos \theta}{r^2 \cos^2 \theta + 1}\end{aligned}$$

(solved)

Q.2 Show that, when f is differentially a function of the point $z = f(xy)$, satisfy the equation

$$x \frac{\delta z}{\delta x} - y \frac{\delta z}{\delta y} = 0$$

Solution: let,

$$z = f(xy) \text{ where, } u = (xy)$$

Now,

$$\frac{\delta z}{\delta x} = \frac{\delta z \delta u}{\delta u \delta x}$$

$$= \frac{\delta z}{\delta u} \frac{\delta}{\delta x}(xy)$$

$$= y \frac{\delta z}{\delta u}$$

Again,

$$\frac{\delta z}{\delta y} = \frac{\delta z \delta u}{\delta u \delta y}$$

$$= \frac{\delta z}{\delta u} \frac{\delta}{\delta y}(xy)$$

$$= x \frac{\delta z}{\delta u}$$

Now,

$$x \frac{\delta z}{\delta x} - y \frac{\delta z}{\delta y}$$

$$= xy \frac{\delta z}{\delta u} - xy \frac{\delta z}{\delta u}$$

$$= 0$$

(solved)

Q.3 Let, f be differential function a f on variable left $z = f(x^2 + y^2)$

Then show that $y \frac{\delta z}{\delta x} - x \frac{\delta z}{\delta y} = 0$

Solution:

Let, $z = f(u)$ where, $u = (x^2 + y^2)$

By chain rule,

$$\begin{aligned}\frac{\delta z}{\delta x} &= \frac{\delta z \delta u}{\delta u \delta x} \\ &= \frac{\delta z}{\delta u} \frac{\delta}{\delta x}(x^2 + y^2) \\ &= 2x \frac{\delta z}{\delta u}\end{aligned}$$

Again,

$$\begin{aligned}\frac{\delta z}{\delta y} &= \frac{\delta z \delta u}{\delta u \delta y} \\ &= \frac{\delta z}{\delta u} \frac{\delta}{\delta y}(x^2 + y^2) \\ &= 2y \frac{\delta z}{\delta u}\end{aligned}$$

Now,

$$\begin{aligned}y \frac{\delta z}{\delta x} - x \frac{\delta z}{\delta y} \\ &= 2xy \frac{\delta z}{\delta u} - 2xy \frac{\delta z}{\delta u} \\ &= 0\end{aligned}$$

(solved)

Q.4 If $z = e^{x^2 y}$, $x = \sqrt{uv}$ and $y = \frac{1}{v}$ then find $\frac{\delta z}{\delta u}$ and $\frac{\delta z}{\delta v}$

Solution:

Given that,

$$z = e^{x^2 y}$$

$$x = \sqrt{uv}$$

$$\text{and } y = \frac{1}{v}$$

By chain rule we know that,

$$\begin{aligned}\frac{\delta z}{\delta u} &= \frac{\delta z \delta x}{\delta x \delta u} + \frac{\delta z \delta y}{\delta y \delta u} \\ &= \frac{\delta}{\delta x}(e^{x^2 y}) \frac{\delta}{\delta u}(\sqrt{uv}) + \frac{\delta}{\delta y}(e^{x^2 y}) \frac{\delta}{\delta u}\left(\frac{1}{v}\right) \\ &= 2xy e^{x^2 y} \frac{v}{2\sqrt{uv}} + x^2 e^{x^2 y} \cdot 0 \\ &= 2\sqrt{uv} \cdot \frac{1}{v} e^{(\sqrt{uv})^{\frac{21}{v}}} \cdot \frac{v}{2\sqrt{uv}} \\ &= 2\sqrt{uv} \cdot \frac{1}{v} \cdot e^{(uv)^{\frac{1}{v}}} \cdot \frac{v}{2\sqrt{uv}} \\ &= e^u\end{aligned}$$

Again by chain rule, we know that,

$$\begin{aligned}\frac{\delta z}{\delta v} &= \frac{\delta z \delta x}{\delta x \delta v} + \frac{\delta z \delta y}{\delta y \delta v} \\ &= \frac{\delta}{\delta x}(e^{x^2 y}) \frac{\delta}{\delta v}(\sqrt{uv}) + \frac{\delta}{\delta y}(e^{x^2 y}) \frac{\delta}{\delta v}\left(\frac{1}{v}\right) \\ &= 2xy e^{x^2 y} \frac{u}{2\sqrt{uv}} + x^2 e^{x^2 y} \left(-\frac{1}{v^2}\right) \\ &= 2\sqrt{uv} \frac{1}{v} \cdot e^{uv - \frac{1}{v}} \frac{u}{2\sqrt{uv}} - uv e^{uv(\frac{1}{v})} \frac{1}{v^2} \\ &= \frac{u}{v} \cdot e^u - \frac{u}{v} \cdot e^u \\ &= 0\end{aligned}$$

(Solved)

